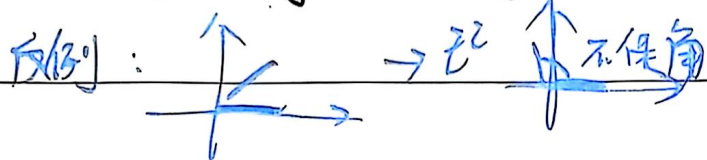


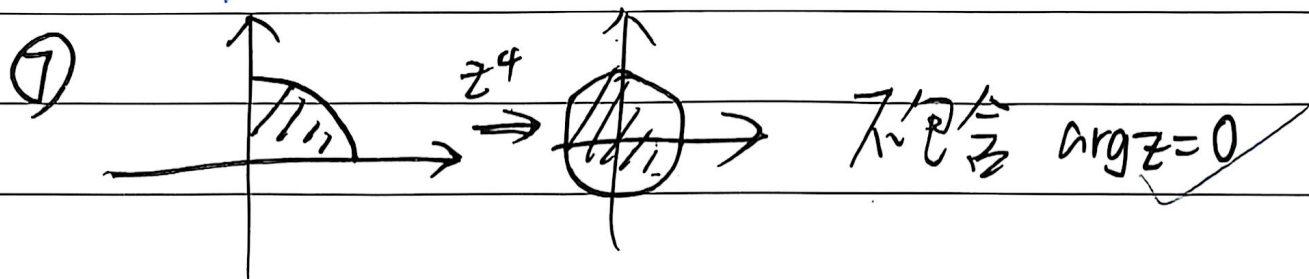
# 思考6

③  $f'(z) = 2z$      $f'(0) = 0$     取  $\Delta z$ ,  $z \rightarrow 0$

则  $\text{Arg} \Delta w - \text{Arg} \Delta z = \text{Arg} \frac{\Delta(z^2)}{\Delta z} = \text{Arg} z$

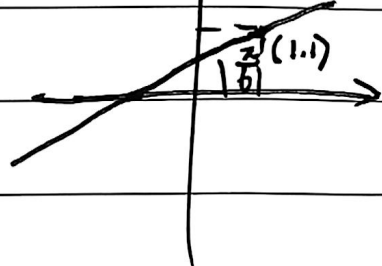


不确定



## 习题6

①  $w = z^2$      $y = \frac{1}{2} + \frac{1}{2}x$      $1+x = 2y \Rightarrow x = 2y - 1$



$w = u + iv = (x + iy)^2 = x^2 - y^2 + 2ixy$

$u = x^2 - y^2$

$v = 2xy$

$\frac{dv}{du} = \frac{d(2xy)}{d(x^2 - y^2)}$

$\frac{dv}{du} = \frac{2d(2y^2 - y)}{d(3y^2 - 4y + 1)}$

~~1. 1~~

$$e^{-y+ic_1} = e^{-y}(\cos c_1 + i \sin c_1)$$

③ 1.  $z = c_1 + iy, u + iv = e^{c_1 + iy} = e^{c_1}(\cos y + i \sin y)$

$$\Rightarrow v = e^{c_1} \sin y$$

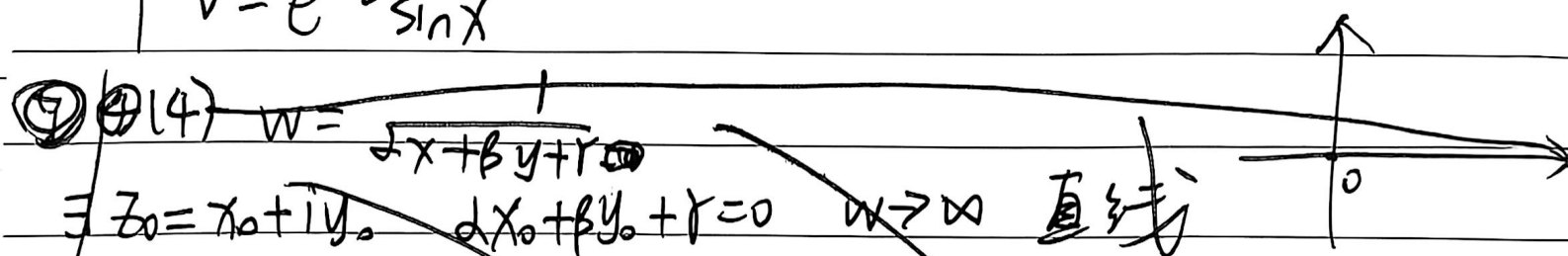
$$u = e^{c_1} \cos y, \therefore v = u \tan c_1, u^2 + v^2 = e^{2c_1} \Rightarrow v = u \tan c_1$$

2.  $z = x + ic_2, u + iv = e^{i(x + ic_2)} = e^{-c_2 + ix} = e^{-c_2}(\cos x + i \sin x)$

$$u = e^{-c_2} \cos x$$

$$u^2 + v^2 = e^{-2c_2}$$

$$v = e^{-c_2} \sin x$$



(5)  $(x+1)^2 + (y+1)^2 = 1$

$z_0 \left\{ \begin{array}{l} x_0 = 1 - \frac{\sqrt{2}}{2} \\ y_0 = 1 - \frac{\sqrt{2}}{2} \end{array} \right.$  时,  $w = \frac{1}{z_0} \rightarrow \infty$  直线

④ (4)  $w = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - i \frac{y}{x^2+y^2}$

$$z = \frac{1}{w} \Rightarrow z = x + iy = \frac{u}{u^2+v^2} - i \frac{v}{u^2+v^2}$$

$$\Rightarrow \frac{\alpha u}{u^2+v^2} - \frac{\beta v}{u^2+v^2} + \gamma = 0 \quad \text{圆}$$

(5)  $(x+1)^2 + (y+1)^2 = 1 \quad w = \frac{1}{x+iy} = \frac{x}{x^2+y^2} - i \frac{y}{x^2+y^2}$

$$z = \frac{1}{w} \Rightarrow z = x + iy = \frac{u}{u^2+v^2} - i \frac{v}{u^2+v^2}$$

$$\left( \frac{u}{u^2+v^2} + 1 \right)^2 + \left( \frac{-v}{u^2+v^2} + 1 \right)^2 = 1$$

$$\frac{u^2}{(u^2+v^2)^2} + \frac{2u-2v}{u^2+v^2} + 1 = 0 \quad \text{圆}$$